

UNIT: 2 INPUT DEVICES

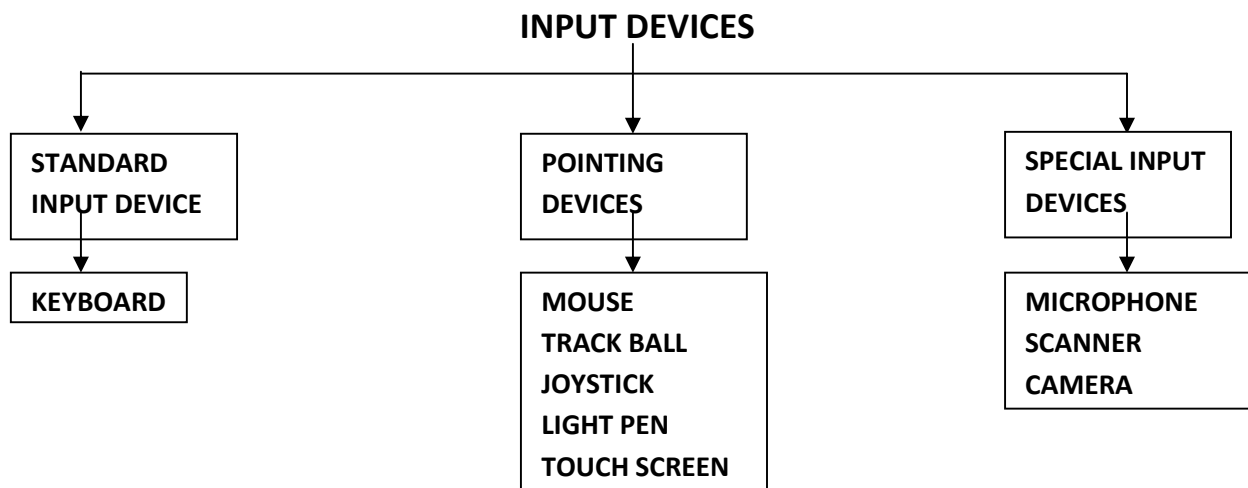
1. WHAT IS INPUT DEVICES?

- The Input devices are the devices which are used to enter the data in the computer system.
- Keyboard, mouse, scanner, microphone are the example of input devices.

FUNCTIONS OF INPUT DEVICES:

- Accept the data from the outside worlds.
- Convert that data into computer coded information.
- Supply this data to Central Processing Unit for further processing.

CLASIFICATION OF INPUT DEVICES:



2. EXPLAIN STANDARD INPUT DEVICE: KEYBOARD.

- Keyboard is most commonly used input device.
- It is similar like a type writer which is used to enter data in the computer.
- It contains sets of keys such as alphabets, number & special signs.
- There are two types of keyboard.
 - General purpose keyboard
 - Special purpose keyboard

GENERAL PURPOSE KEYBOARD:

- Standard keyboard which are used in personal computers.
- It contains enough keys which are used in all types of applications so they are known as general purpose keyboard.

- Most popular general purpose keyboard contains 101 keys.

- The general purpose keyboard are divided into following parts:

ALPHANUMERIC KEYPAD

- The centred part of the keyboard is known as alphanumeric keypad.
- It contains alphabets, numbers & special signs such as *,!, @, #, \$, %,*,* etc.

NUMERIC KEYPAD

- The right most part of the keyboard is known as numeric keypad.
- It contains 0 to 9 numbers & mathematical signs such as +, *, -, /.
- Mainly used for fast data entry in mathematical applications.

ARROW KEYS

- Set of four keys up, down, left & right.
- Used to move the cursor at left & right or up and down on the screen.
- They are referred as “cursor-control” or “cursor-movement” keys.

FUNCTION KEYS

- The first line of the keyboard contains a Set of 12 keys with name f1 to f2 are known as function keys.
- Used to generate short-cuts in different software package.

SPECIAL KEYS

- There are lots of keys that are used for some specific task describes follows:
- TAB: used for gives multiple spaces or move the cursor to next defined position.
- ENTER: used for generate the output of any command.
- SPACE: used to make one blank space between two words.
- BACKSPACE: used to remove the left-most character at cursor position.
- DELETE: used to remove the right-most character at cursor position.
- HOME: moves the cursor at the beginning of the line.
- END: moves cursor at the end of the line.
- PAGE UP: moves or scroll the screen up or previous page of the current page.
- PAGE DOWN: moves the screen to the next page from the currently displayed page.
- PRINT SCREEN: used to print what is currently displayed on the screen.
- INSERT: used to enter text between two characters.

- **ESC:** used to negate current command or terminate the execution of the program.
- **ALT:** used to expand the functionality of keyboard. Basically used to generate shortcuts in different application.
- **CTRL:** used to expand the functionality of keyboard. Basically used to generate shortcuts in different application.
- **NUMLOCK:** used to on or off the numeric keypad.
- **CAPSLOCK:** used to type the all inputted text capitally.

SPECIAL PURPOSE KEYBOARD

- Special purpose keyboard is used for special purpose applications which required faster data entry and rapid interaction with the computer system.
- For example ATM used in banks used special purpose keyboard which contains a few keys.

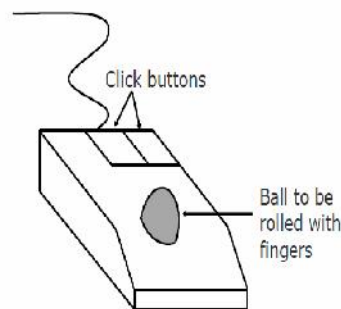
3. EXPLAIN POINTING DEVICES.

1. MOUSE

- Mouse is Small hand-hold device Input device which is generally used for drawing purpose.
- It's a Pointing device.
- It contains two or three buttons
- Left button is used to point out or select any item by clicking.
- Right to generate context menu.
- When user moves mouse across flat surface, the graphic cursor moves on screen.
- Graphic cursor contains verity of symbols such as arrow, wrist, pointing finger etc.
- Depending on application text & graphic cursors are changed.
- The following 5 techniques are used to carry out various operations:
- **POINT:**
 - To move the mouse on top of icon
- **C CLICK:**
 - To press & release the left button of mouse at once.
 - Used to open any currently selected icon, menu.
- **DOUBLE CLICK:**
 - To press & release the left button of mouse twice.
 - Used to open any application or program.

- **SIMULTANEOUS-CLICK:**
 - Press & release left & right button to gather.
 - Used in some software package to added some functionality.
- **DRAG:**
 - Press the left button down & moved the mouse on screen.
 - Used to move the graphics on screen.
- Many types of mouse are available such as mechanical mouse, optical mouse, serial mouse, wireless mouse which are used for different purpose.

2. TRACK BALL



Commonly used in laptop (notebook) computers

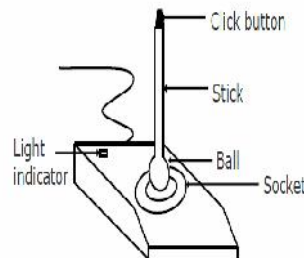
- Trackball is a pointing device which is similar to a mouse.
- A ball is placed on the track ball device which is used to move the graphic cursor on the screen.
- It also contains buttons which are used to select a particular item on the screen.
- To move the graphic cursor on screen, the ball is rolled with the fingers or thumb.
- It needs not to move the whole device to move the cursor so it is often attached with some keyboards.
- Track balls come in various shapes with same functionality.
- Commonly three shapes are used: A BALL, A SQUARE, and A SQUARE.
- In case of ball we need to move it with the help of finger.
- In case of button pushed with finger in desired direction of the cursor movement.
- In case of button press finger to up or down & left or right to move cursor.

Advantages of track ball

- Takes less desk space.

- Takes less arm movements than mouse.
- Doesn't require any mouse pad & large area to move the mouse.
- Less strain on the wrist.
- Finger trip control which may offer more accuracy than mouse.

3. JOYSTICK

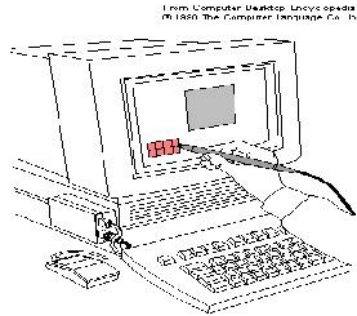


Commonly used for video games, flight simulators, training simulators, and for controlling industrial robots

- Joystick is a pointing device which works on the same principle of track ball.
- It contains a stick which is placed on the spherical ball.
- The stick is used to move the cursor at desired position left or right or backward or forward.
- It also contains a button that is clicked to make selection of currently pointed item.
- A joystick is similar to a mouse, except that with a mouse the cursor stops moving as soon as you stop moving the mouse.
- With a joystick, the pointer continues moving in the direction the joystick is pointing.
- To stop the pointer, you must return the joystick to its upright position.
- Some of the systems using joysticks are
 - Aircrafts, UAVs for flight control
 - Motorized Wheelchairs as input device
 - Microscopes
 - Submarines
 - Security Systems
 - Video Games
- Joysticks are widely used for video games
- Advantages of joystick
- It is very easy to learn to use.

- Very simple design so they can be inexpensive.
- It has a big analogue stick in the middle so it's easier to control.

4. LIGHT PEN



- Light pen is a pointing device which is used to draw directly on the screen.
- It is called light pen because it is similar to a pen & senses light.
- It's an input device in the form of light-sensitive stick used in conjunction with a CRT display.
- The light pen allows the user to point out or draw any object on the screen.
- The user brings the pen to the desired point on screen and presses the pen button to make contact.
- It has a switch on its top which allows the user to make contact with screen.
- It is useful for drawing or graphics in the program such as CAD (computer aided design).
- An engineer, architect or fashion designer can draw directly on screen.
- Used in application such as gaming, graphic arts, healthcare applications etc.
- Light pen cannot scratch or damage a screen.

Advantages of light pen;

- Less expensive than touch screen.
- give the user the full range of mouse capabilities, without the use of a pad or any horizontal surface
- Cannot scratch or damage screen.
- Works on any size screen.

5. TOUCH SCREEN

- Touch screen is a pointing device.
- It is most simple & easiest to learn of all input devices.

- It allows the user to choose from available options by simply touching with their figure to the desired icon or menu item displayed on the computer screen.
- A touch screen is an electronic visual display that can detect the presence and location of a touch within the display area.
- The term generally refers to touching the display of the device with a finger or hand.
- Touch screens are common in devices such as computers, tablet computers & Smartphone.
- The touch screen has two main attributes::
 - First, it enables one to interact directly with what is displayed.
 - Secondly, it lets one do so without requiring any intermediate device that would need to be held in the hand
- It's a very easy to operate device which users can use the system without any formal training.
- Uses optical sensors that detect the touch of the finger on screen.
- Sensors communicate the position of touch to the computer which interprets the input made by the users.
- It contains pressure sensitive monitors which are placed inside the base of computer screen.
- Pressure sensitive monitors contain sensors to measure the monitor's weight at many points.
- When user touches the screen, the changes on weights & forces transferred down to sensor which allows the device to detect the location of the touch.
- This type of monitors required little pressure to transmit the desired input.
- Touch screen are commonly used in following places.
 - An airport or railway station.
 - Large departmental stores.
 - In large museums or zoos to guide visitors to the locations of various attractions.
 - Self service check out
 - In ATM machines
 - In I-phones or PDA's
 - Computer based training
- Any type of touch screen contains Three main components:

A touch screen sensor panel:

- Which sits above the display & generate appropriate voltage according to where precisely it is touched?

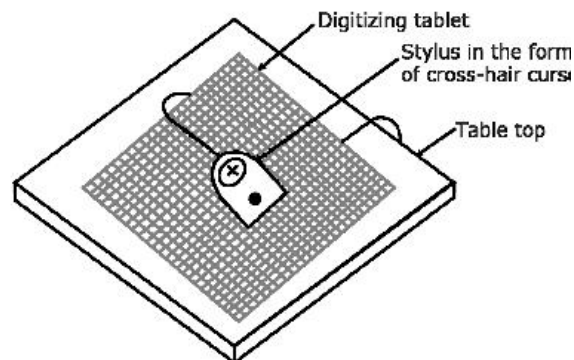
A touch screen controller:

- Processes the signal received from the sensor & translates this touch event data & passed to pc's processor via serial or USB interface.

A soft ware driver:

- Provides an interface to the pc's operating system & which translates the touch event data into mouse event.

6. DIGITIZER



- An input device.
- Used for converting pictures, map & drawing into digital form.
- Allows one to hand-draw images and graphics, similar to the way one draws images with a pencil and paper.
- Also be used to capture data or handwritten signatures.
- The device consists of a flat surface upon which the user may "draw" an image using an attached stylus, a pen-like drawing tool.
- These devices are usually connected via a Serial port.
- Placed on the desk n connected with the computer.
- Digitizer consists of graphic tablets which are associated with a stylus.
- The stylus is like a pen with a button.
- Stylus connected with a tablet and can press down at a point on the tablet to input (x, y) co-ordinates of point.
- It contains hundreds of copper wires forming a grid that receives electric pulsed.
- When stylus moves on tables the cursor on screen moves simultaneously.
- Allows the user to draw sketches directly.
- Commonly used in CAD by architects & engineers.

- Used in GIS (geographical information system) for digitizing maps.

7. MICROPHONE

- It's an input device.
- Used to store the voice data into the computer system.
- Microphones are a type of *transducer* - a device which converts energy from one form to another.
- Microphones convert sound waves into electrical energy.
- Different types of microphone have different ways of converting energy.



- All the microphones share one common thing: The diaphragm.
- Thin piece of material (such as paper, plastic or aluminium) which vibrates when it is struck by sound waves.
- When the diaphragm vibrates, it causes other components in the microphone to vibrate.
- These vibrations are converted into an electrical current which becomes the audio signal.
- The microphones are divided into mainly two types

The type of conversion technology they use

- This refers to the technical method the mike uses to convert sound into electricity.
- The most common technologies are dynamic, condenser, ribbon and crystal.

The type of application they are designed for

- Some mikes are designed for general use and others are much specialised purpose.

8. WEB CAMERA

- It's an input device.
- Used to feed the image to a computer or computer network often via USB or Wi-Fi.

- Web camera is a hardware camera connected to a computer that allows everyone to connect to internet to view either pictures or motion video.
- Most Web cameras are embedded to display with laptop computer or connected with USB or Wi-Fi with a computer.
- Simple web cam. Consists a digital camera attached to your computer typically through USB.
- The camera part of web camera is just a digital camera.
- Web camera comes with software which preset interval & transfer it to another location of viewing.
- Web camera system allows you to using video also for that you have web camera with high frame rate.
- Web camera is a digital camera which taking picture over & over & again one after another.
- These images are stored image into the physical memory of camera in built in.
- After capture image & stored in memory it reduced the amount of data need to transmit.
- Web camera software takes image & converts data in jpeg (compressing format).

9. EXPLAIN SCANNER & ITS TYPES.

- Scanners are input devices.
- They are capable of entering information directly into the computer.
- The main advantage of direct entry of information is that users do not have to key the information.
- And another advantage is that through Scanners you can input Graphical Data into the computer. This
- Provides faster and more accurate data entry. Important types of scanners

Image scanner:

- It's an input device, which translates paper documents into an electronic format which can be stored in a computer.
- The input document may be typed text, pictures, graphics or even handwritten material.
- There are two types of image scanner:

Flatbed scanner

- It's like a xerox machine which consist of box having a glass plate on its top and a lid to covers the glass plate.
- The document placed inside the glass plate & light source is situated below glass plate which moves horizontally from left to right & scanning document line by line.

Handheld scanner

- It contains a set of light emitting diodes encased in small case which can be conveniently held in hand.
- To scan a document the scanner is slowly dragged on the document.
- The scanner has to be dragged carefully & steadily otherwise the document cannot scan properly.
- Used when higher accuracy is not required.

10. EXPLAIN THE OPTICAL SCANNERS

OMR (OPTICAL MARK READER)

- OMR is a device that is capable of recognised pre-specified type of mark made by pencil or pen.
- The Optical Mark Reader is a device which can detect the presence or absence of a mark on a paper.
- The OMR recognise the marks by focusing a light on the paper being scanned & detect the reflected light pattern from mark.
- The present mark is detected due to intensity of light being reflected from the mark.
- Pencil marks made with soft lead pencil reflect the light which allowing the OMR to determine which response are mark.
- OMR is used in reading answers sheets, questionnaires.

ADVANTAGES

- Speedy and accurate to generate result.
- Cheap in cost.

DISADVANTAGE

- Cannot able to read characters.
- Erasing or cancellation is not possible.
- Good quality expensive paper is required.

OCR (OPTICAL CHARACTER READER) DEVICE

- OCR capable of recognizing alphabets & numbers printed on paper.
- It can also capable of recognise shape & identify character directly from source document.
- It is always used with character recognized software.
- It converts bitmap images of character to equivalent ASCII code.
- First it create bitmap image of document & OCR software translate into ASCII code which computer interprets letter, symbol or number.
- The type of document must be type using OCR fonts.
- The software design to recognised the standard OCR-A(American standard) & OCR-B(European standard)

ADVANTAGES

- Speedy entered data.
- Accept wide range of font using ordinary mark.

DISADVANTAGE

- Expensive
- Scanned properly only if the characters are standard size.
- Dusty paper cant scanned properly.

MICR (MAGNETIC INK CHARACTER RECOGNITION)

- MICR systems use special ink which can be magnetized, to print characters that can then be read and decoded by special magnetic devices.
- The common E13B font is used to write these special kinds of cheques.
- E13B font contains 0-9 numbers & 4 symbols.
- Detection of characters is a two step process.
- First MICR Reader-Sorter reads the data on cheques & sorts the cheques for distribution for further processing.
- The reading station is used to sense and identify the magnetic characters as they pass through.
- Magnetized characters are read by the head. E.g. MICR is used in Banks to cheques the MICR systems use special ink which can be magnetized, to print characters cheques.

ADVANTAGES

- Speedy data entry.
- Accurate output.
- Folded or roughly handled cheques are also scanned with same accuracy.

DISADVANTAGE

- MICR software is required.
- Limited fonts are used so used in banking industries only.

BCR (BAR CODE READER) DEVICE

- Data can be coded in the form of small lines which are known as Bar Codes.
- Bar codes represent the alphanumeric data by combination of vertical lines which contains different width & spacing between them.
- Bar Code Reader is a device which are used to recognized bar code data.
- It scanned the barcode image & converted into alphanumeric value & fed to computer.
- It uses laser-beam technology. Laser stroke across the pattern of bar which sensed by light sensitive decoder & their reflection of light pattern are converted into electrical pulse which converted it into alphanumeric value.
- Various barcodes are available for different use.
- Most common is UPC (UNIVERSAL PRODUCT CODE).

It contain 10 digit first 5 identify the manufacturer name & remaining identify a specific product.